

Chapter two

1. Channel

In telecommunications and computer networking, a communication channel refers either to a physical transmission medium, such as a wire, or to a logical connection over a multiplexed medium, such as a radio channel. A channel is used to convey an information signal, for example, a digital bit stream, from one or several senders (or transmitters) to one or several receivers. A channel has a certain capacity for transmitting information, often measured by its bandwidth in Hz or its data rate in bits per second.

2. Symmetric channel:

The symmetric channel has the following conditions:

a- Equal number of symbols in X&Y, i.e. $P(Y|X)$ is a square matrix.

b- Any row in the $P(Y|X)$ matrix comes from some permutation of the other rows.

For example, the following conditional probability of various channel types as shown:

a- $P(Y|X) = \begin{bmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{bmatrix}$ is a BSC, because it is square matrix and 1st row is the permutation of 2nd row.

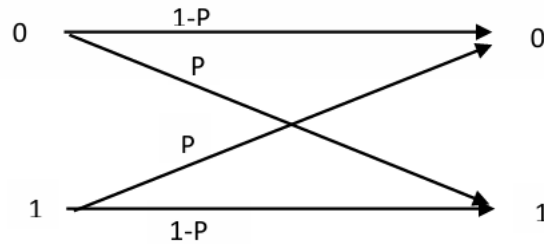
b- $P(Y|X) = \begin{bmatrix} 0.9 & 0.05 & 0.05 \\ 0.05 & 0.9 & 0.05 \\ 0.05 & 0.05 & 0.9 \end{bmatrix}$ It is TSC because it is a square matrix, and each row is a permutation of the others.

c- $P(Y|X) = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.8 & 0.1 \end{bmatrix}$ is a non-symmetric since since it is not square although each row is permutation of others.

d- $P(Y|X) = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.1 & 0.1 & 0.8 \end{bmatrix}$ is a non-symmetric although it is square since 2nd row is not permutation of other rows.

2.1.Binary symmetric channel (BSC)

It is a common communications channel model used in coding theory and information theory. In this model, a transmitter wishes to send a bit (a zero or a one), and the receiver receives a bit. It is assumed that the bit is usually transmitted correctly, but that it will be "flipped" with a small probability (the "crossover probability").



A binary symmetric channel with crossover probability p denoted by BSC_p , is a channel with binary input and binary output and probability of error p ; that is, if X is the transmitted random variable and Y the received variable, then the channel is characterized by the conditional probabilities:

$$Pr(Y = 0 \mid X = 0) = 1 - P$$

$$Pr(Y = 0 \mid X = 1) = P$$

$$Pr(Y = 1 \mid X = 0) = P$$

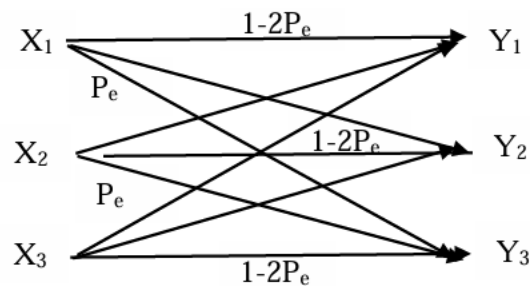
$$Pr(Y = 1 \mid X = 1) = 1 - P$$

2.2. Ternary symmetric channel (TSC)

The transitional probability of TSC is:

$$P(Y|X) = \begin{bmatrix} 1 - 2P_e & P_e & P_e \\ P_e & 1 - 2P_e & P_e \\ P_e & P_e & 1 - 2P_e \end{bmatrix}$$

The TSC is symmetric but not very practical, since in practice x_1 and x_3 are not affected as much as x_2 . In fact, the interference between x_1 and x_3 is much less than the interference between x_1 and x_2 or x_2 and x_3 .



Hence, the more practice, the more nonsymmetric the channel's transitional probability becomes.

$$P(Y|X) = \begin{bmatrix} 1 - P_e & P_e & 0 \\ P_e & 1 - 2P_e & P_e \\ 0 & P_e & 1 - P_e \end{bmatrix}$$

3. Shannon's theorem

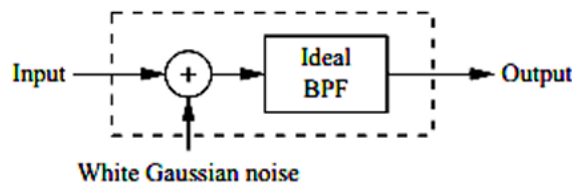
a- A given communication system has a maximum rate of information C known as the channel capacity.

b- If the information rate R is less than C , then one can approach arbitrarily small error probabilities by using intelligent coding techniques.

c- To get lower error probabilities, the encoder has to work on longer blocks of signal data. This entails longer delays and higher computational requirements.

Thus, if $R \leq C$ then transmission may be accomplished without error in the presence of noise. The negation of this theorem is also true: if $R > C$, then errors cannot be avoided regardless of the coding technique used.

Consider a bandlimited Gaussian channel operating in the presence of additive Gaussian noise:



The Shannon-Hartley theorem states that the channel capacity is given by:

$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

Where C is the capacity in bits per second, B is the bandwidth of the channel in Hertz, and S/N is the signal-to-noise ratio.

Example 1: A channel having a bandwidth of 1 kHz, the power of signal is 12 W and the power of noise is 4 W. Determine the channel capacity.

Solution:

$$C = B \log_2 \left(1 + \frac{S}{N} \right) = 1 \times 10^3 \log_2 \left(1 + \frac{12}{4} \right) = 2 \text{ bps.}$$

Or: A channel having a bandwidth of 1 kHz and Signal to Noise Ratio (SNR) is 3. Determine the channel capacity.

Example 2: If the channel capacity is 2000 bps, find its bandwidth if the SNR is 3.

Solution:

$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

$$2000 = B \log_2(1 + 3) = B \log_2(4)$$

$$2000 = B \times 2 \rightarrow B = 1000 \text{ kHz}$$

Example 3: Determine the SNR required to achieve a channel capacity of 1585 bps, if the bandwidth is 1 kHz.

Solution:

$$C = B \log_2(1 + \text{SNR})$$

$$1585 = 1000 \log_2(1 + \text{SNR})$$

$$\frac{1585}{1000} = \log_2(1 + \text{SNR})$$

$$1.585 = \log_2(1 + \text{SNR})$$

$$\log_2^{-1}(1.585) = (1 + \text{SNR})$$

$$3 = (1 + \text{SNR})$$

$$\text{SNR} = 3 - 1 = 2$$

4. Channel Capacity (Discrete channel)

This is defined as the maximum of $I(X, Y)$:

$$C = \text{channel capacity} = \max[I(x, y)] \quad \frac{\text{bits}}{\text{symbol}}$$

Physically it is the maximum amount of information each symbol can carry to the receiver. Sometimes this capacity is also expressed in bits/sec if related to the rate of producing symbols r :

$$R(X, Y) = r \times I(X, Y) \text{ bits/sec or } R(X, Y) = I(X, Y) / \bar{\tau}$$

Channel capacity of Symmetric channels:

The channel capacity is defined as $\max [I(X, Y)]$

$$I(X, Y) = H(Y) - H(Y | X)$$

$$I(X, Y) = H(Y) + \sum_{j=1}^m \sum_{i=1}^n P(x_i, y_j) \log_2(y_j | x_i)$$

But we have

$$P(x_i, y_j) = P(x_i)P(y_j | x_i) \quad \text{put in above equation yields:}$$

$$I(X, Y) = H(Y) + \sum_{j=1}^m \sum_{i=1}^n P(x_i)P(y_j | x_i) \log_2 P(y_j | x_i)$$

If the channel is symmetric the quantity:

$$\sum_{j=1}^m P(y_j | x_i) \log_2 P(y_j | x_i) = K$$

Where K is constant and independent of the row number i so that the equation becomes:

$$I(X, Y) = H(Y) + K \sum_{i=1}^n P(x_i)$$

Hence $I(X, Y) = H(Y) + K$ for symmetric channels

$$\text{Max of } I(X, Y) = \max[H(Y) + K] = \max[H(Y)] + K$$

When Y has equiprobable symbols then $\max[H(Y)] = \log_2 m$

Then

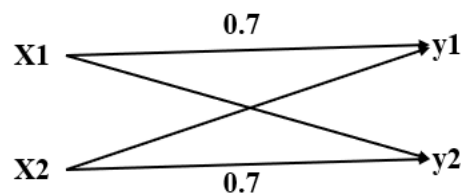
Then

$$5. I(X, Y) = \log_2 m + K$$

Or

$$C = \log_2 m + K$$

Example 4: For the BSC shown below:



Find the channel capacity and efficiency if $I(x_1) = 2 \text{ bits}$.

Solution:

$$P(Y | X) = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix}$$

Since the channel is symmetric then

$$C = \log_2 m + K \quad \text{and } n = m$$

where n and m are number row and column respectively

$$K = 0.7 \log_2 0.7 + 0.3 \log_2 0.3 = -0.88129$$

$$C = 1 - 0.88129 = 0.1187 \text{ bits/symbol}$$

The channel efficiency $\eta = \frac{I(X,Y)}{C}$

$$I(x_1) = -\log_2 P(x_1) = 2$$

$$P(x_1) = 2^{-2} = 0.25 \quad \text{then } P(X) = [0.25 \quad 0.75]^T$$

And we have $P(x_i, y_j) = P(x_i)P(y_j | x_i)$ so that

$$P(X, Y) = \begin{bmatrix} 0.7 \times 0.25 & 0.3 \times 0.25 \\ 0.3 \times 0.75 & 0.7 \times 0.75 \end{bmatrix} = \begin{bmatrix} 0.175 & 0.075 \\ 0.225 & 0.525 \end{bmatrix}$$

$$I(Y) = [0.4 \ 0.6] \rightarrow H(Y) = 0.97095 \frac{\text{bits}}{\text{symbol}}$$

$$I(X, Y) = H(Y) + K = 0.97095 - 0.88129 = 0.0896 \text{ bits/symbol}$$

$$\text{Then } \eta = \frac{0.0896}{0.1187} = 75.6\%$$

To find the channel redundancy:

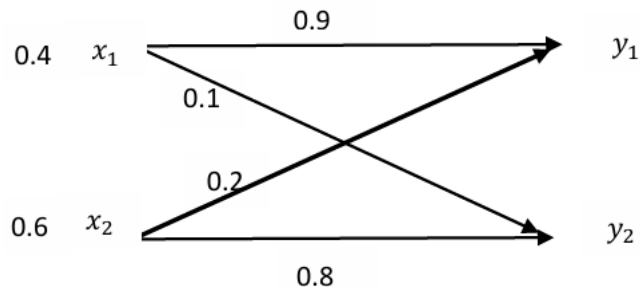
$$R = 1 - \eta = 1 - 0.756 = 0.244 \text{ or } 24.4\%$$

Example 5: A binary source sending x_1 with a probability of 0.4 and x_2 with 0.6 probability through a channel with a probabilities of errors of 0.1 for x_1 and 0.2 for x_2 . Determine:

- 1- Source entropy.
- 2- Marginal entropy.
- 3- Joint entropy.
- 4- Conditional entropy $H(Y | X)$.
- 5- Losses entropy $H(X | Y)$.
- 6- Transinformation.

Solution:

1- The channel diagram:



$$\text{Or } P(Y|X) = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

$$H(X) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i)$$

$$H(X) = - \frac{[0.4 \ln(0.4) + 0.6 \ln(0.6)]}{\ln 2} = 0.971 \frac{\text{bits}}{\text{symbol}}$$

$$2- P(X,Y) = P(Y|X) \times P(X)$$

$$P(Y|X) = \begin{bmatrix} 0.9 \times 0.4 & 0.1 \times 0.4 \\ 0.2 \times 0.6 & 0.8 \times 0.6 \end{bmatrix} = \begin{bmatrix} 0.36 & 0.04 \\ 0.12 & 0.48 \end{bmatrix}$$

$$H(Y) = - \sum_{j=1}^m P(y_j) \log_2 P(y_j) \quad \frac{\text{bits}}{\text{symbol}}$$

$$H(Y) = 0.999 \frac{\text{bits}}{\text{symbols}}$$

$$3- H(X,Y)$$

$$H(X,Y) = - \sum_{j=1}^m \sum_{i=1}^n P(x_i, y_j) \log_2 P(x_i, y_j)$$

$$H(X,Y) = 1.592 \text{ bits/symbol}$$

$$4- H(Y|X)$$

$$\begin{aligned}
 H(Y | X) &= - \sum_{j=1}^m \sum_{i=1}^n P(x_i, y_j) \log_2 P(y_j | x_i) \quad \text{bits/symbol} \\
 &= - \frac{[0.36 \ln(0.9) + 0.12 \ln(0.2) + 0.04 \ln(0.1) + 0.48 \ln(0.8)]}{\ln(2)} \\
 &= 0.621 \frac{\text{bits}}{\text{symbol}}
 \end{aligned}$$

$$\text{Or } H(Y | X) = H(X, Y) - H(X) = 1.592 - 0.971 = 0.621 \frac{\text{bits}}{\text{symbol}}$$

$$5- H(X | Y) = H(X, Y) - H(Y) = 1.592 - 0.999 = 0.593 \text{ bits/symbol}$$

$$6- I(X, Y) = H(X) - H(X | Y) = 0.971 - 0.593 = 0.378 \text{ bits/symbol}$$

6. Cascading of Channels

If two channels are cascaded, then the overall transition matrix is the product of the two transition matrices.

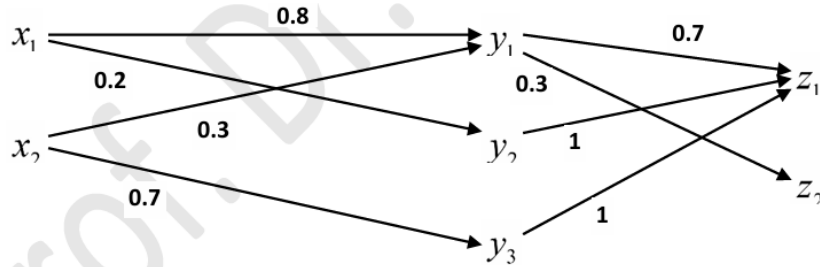
$$p(z/x) = p(y/x) \cdot p(z/y)$$

$$\begin{matrix} (n \times k) & (n \times m) & (m \times k) \\ \text{matrix} & \text{matrix} & \text{matrix} \end{matrix}$$

For the series information channel, the overall channel capacity does not exceed any of the channels individually.

$$I(X, Z) \leq I(X, Y) \text{ \& } I(X, Z) \leq I(Y, Z)$$

Example 6: Find the transition matrix for the cascaded channel shown.



Solution:

$$p(y/x) = \begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.3 & 0 & 0.7 \end{bmatrix}, \quad p(z/y) = \begin{bmatrix} 0.7 & 0.3 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$p(z/x) = \begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.3 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0.76 & 0.24 \\ 0.91 & 0.09 \end{bmatrix}$$